

Aansh Jha Homework 5

Load the Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

csv_file = 'data/nyccrashes_2024w0630_by20240916.csv'
feather_file = 'data/nyccrashes_cleaned.feather'

df = pd.read_csv(csv_file)

df.to_feather(feather_file)

csv_size = os.path.getsize(csv_file)
feather_size = os.path.getsize(feather_file)

csv_size_mb = csv_size / (1024 * 1024)
feather_size_mb = feather_size / (1024 * 1024)

print(f"CSV file size: {csv_size_mb:.2f} MB")
print(f"Feather file size: {feather_size_mb:.2f} MB")

dff = pd.read_feather(feather_file)
print(dff.shape)
print(dff.head())

print(df.columns)
```

CSV file size: 0.34 MB

Feather file size: 0.19 MB

(1875, 29)

	CRASH DATE	CRASH TIME	BOROUGH	ZIP CODE	LATITUDE	LONGITUDE	\
0	06/30/2024	17:30	None	NaN	0.00000	0.00000	
1	06/30/2024	0:32	None	NaN	NaN	NaN	
2	06/30/2024	7:05	BROOKLYN	11235.0	40.58106	-73.96744	
3	06/30/2024	20:47	None	NaN	40.76363	-73.95330	
4	06/30/2024	10:14	BROOKLYN	11222.0	40.73046	-73.95149	

	LOCATION	ON STREET NAME	CROSS STREET NAME	\
0	(0.0, 0.0)	None	None	
1	None	BELT PARKWAY RAMP	None	
2	(40.58106, -73.96744)	None	None	
3	(40.76363, -73.9533)	FDR DRIVE	None	
4	(40.73046, -73.95149)	GREENPOINT AVENUE	MC GUINNESS BOULEVARD	

	OFF STREET NAME	...	CONTRIBUTING FACTOR VEHICLE 2	\
0	GOLD STREET	...	Unspecified	
1	None	...	Unspecified	
2	2797 OCEAN PARKWAY	...	None	
3	None	...	None	
4	None	...	Unspecified	

	CONTRIBUTING FACTOR VEHICLE 3	CONTRIBUTING FACTOR VEHICLE 4	\
0	None	None	
1	None	None	
2	None	None	
3	None	None	
4	None	None	

	CONTRIBUTING FACTOR VEHICLE 5	COLLISION_ID	\
0	None	4736746	
1	None	4736768	
2	None	4737060	
3	None	4737510	
4	None	4736759	

	VEHICLE TYPE CODE 1	VEHICLE TYPE CODE 2	\
0	Sedan	Sedan	
1	Station Wagon/Sport Utility Vehicle	Station Wagon/Sport Utility Vehicle	
2	Station Wagon/Sport Utility Vehicle	None	
3	Sedan	None	

4

Bus

Box Truck

	VEHICLE TYPE CODE 3	VEHICLE TYPE CODE 4	VEHICLE TYPE CODE 5
0	None	None	None
1	None	None	None
2	None	None	None
3	None	None	None
4	None	None	None

```
[5 rows x 29 columns]
```

```
Index(['CRASH DATE', 'CRASH TIME', 'BOROUGH', 'ZIP CODE', 'LATITUDE',
      'LONGITUDE', 'LOCATION', 'ON STREET NAME', 'CROSS STREET NAME',
      'OFF STREET NAME', 'NUMBER OF PERSONS INJURED',
      'NUMBER OF PERSONS KILLED', 'NUMBER OF PEDESTRIANS INJURED',
      'NUMBER OF PEDESTRIANS KILLED', 'NUMBER OF CYCLIST INJURED',
      'NUMBER OF CYCLIST KILLED', 'NUMBER OF MOTORIST INJURED',
      'NUMBER OF MOTORIST KILLED', 'CONTRIBUTING FACTOR VEHICLE 1',
      'CONTRIBUTING FACTOR VEHICLE 2', 'CONTRIBUTING FACTOR VEHICLE 3',
      'CONTRIBUTING FACTOR VEHICLE 4', 'CONTRIBUTING FACTOR VEHICLE 5',
      'COLLISION_ID', 'VEHICLE TYPE CODE 1', 'VEHICLE TYPE CODE 2',
      'VEHICLE TYPE CODE 3', 'VEHICLE TYPE CODE 4', 'VEHICLE TYPE CODE 5'],
      dtype='object')
```

1. Construct a Contingency Table for Missing Geocode Data by Borough

```
import pandas as pd
import scipy.stats as stats

df['Missing_Geocode'] = df['LATITUDE'].isnull() | df['LONGITUDE'].isnull()

# Create the contingency table
contingency_table = pd.crosstab(df['BOROUGH'], df['Missing_Geocode'])
print(contingency_table)

# Chi-square test
chi2, p, dof, expected = stats.chi2_contingency(contingency_table)
print(f"Chi-Square Statistic: {chi2}, P-value: {p}")

if p < 0.05:
```

```

    print("Reject the null hypothesis: Missing geocode data is associated with the borough.")
else:
    print("Fail to reject the null hypothesis: No association between missing geocode data and borough.")

```

Missing_Geocode	False	True
BOROUGH		
BRONX	208	5
BROOKLYN	455	7
MANHATTAN	221	7
QUEENS	375	6
STATEN ISLAND	48	2

Chi-Square Statistic: 3.346164674960958, P-value: 0.5016512622580913

Fail to reject the null hypothesis: No association between missing geocode data and borough.

2. Create Hour Variable and Plot Histogram of Crashes by Hour

```

import matplotlib.pyplot as plt

# Convert CRASH TIME to datetime and extract hour
df['CRASH TIME'] = pd.to_datetime(df['CRASH TIME'], format='%H:%M')
df['hour'] = df['CRASH TIME'].dt.hour

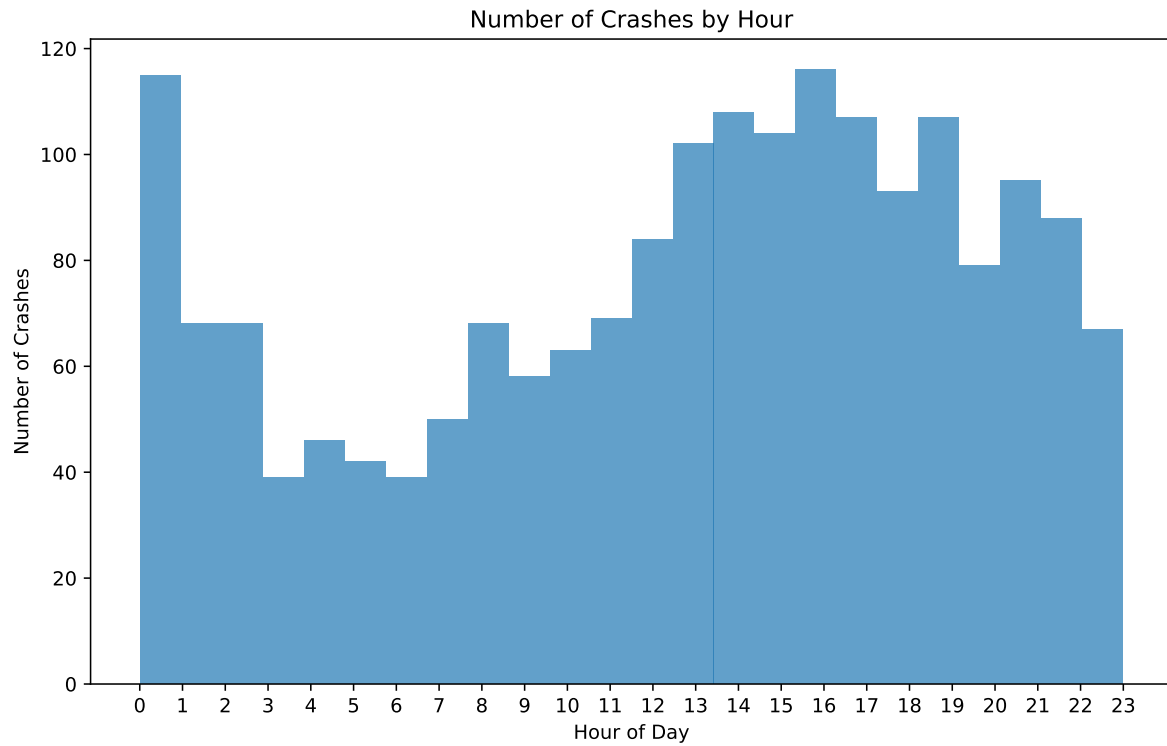
# Plot histogram for all data
plt.figure(figsize=(10, 6))
plt.hist(df['hour'], bins=24, alpha=0.7)
plt.xlabel('Hour of Day')
plt.ylabel('Number of Crashes')
plt.title('Number of Crashes by Hour')
plt.xticks(range(24))
plt.show()

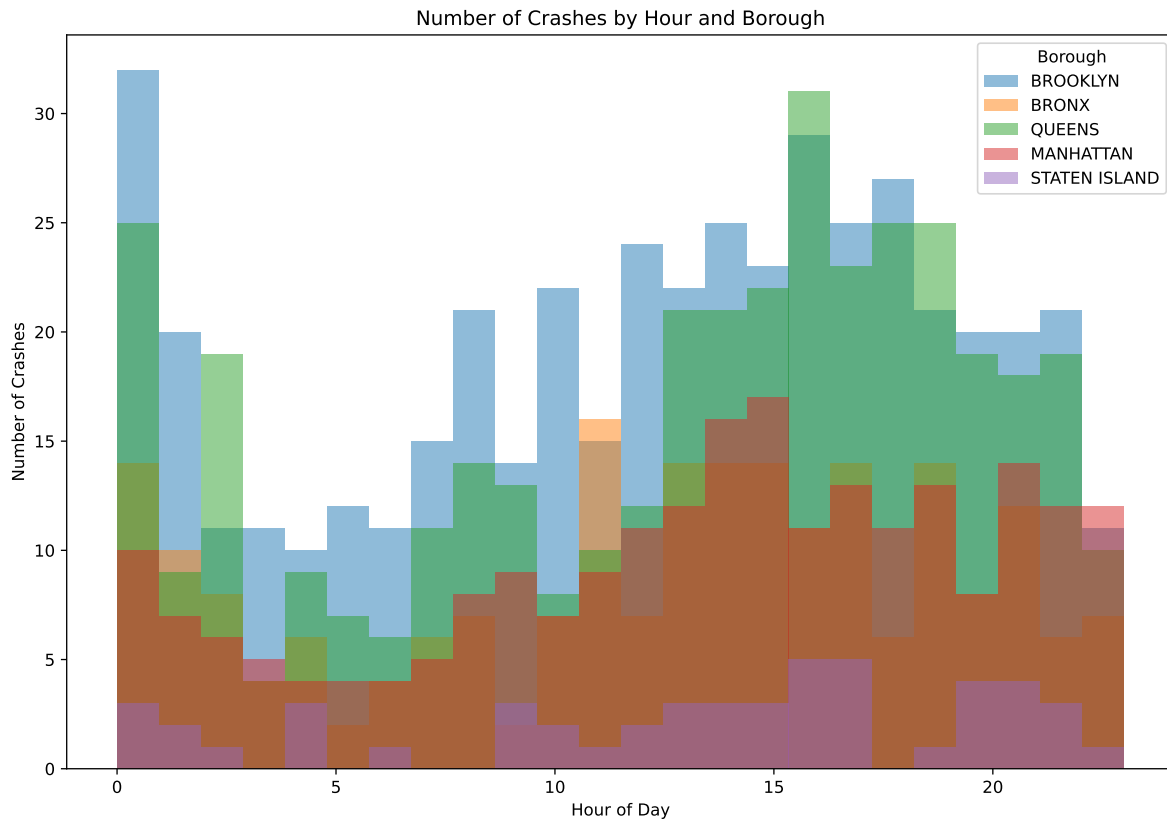
# Plot histogram for each borough
plt.figure(figsize=(12, 8))
for borough in df['BOROUGH'].dropna().unique():
    subset = df[df['BOROUGH'] == borough]
    plt.hist(subset['hour'], bins=24, alpha=0.5, label=borough)

plt.xlabel('Hour of Day')
plt.ylabel('Number of Crashes')

```

```
plt.title('Number of Crashes by Hour and Borough')  
plt.legend(title='Borough')  
plt.show()
```





3. Overlay Crash Locations on a Map of NYC

```
import geopandas as gpd
from shapely.geometry import Point
import contextily as ctx

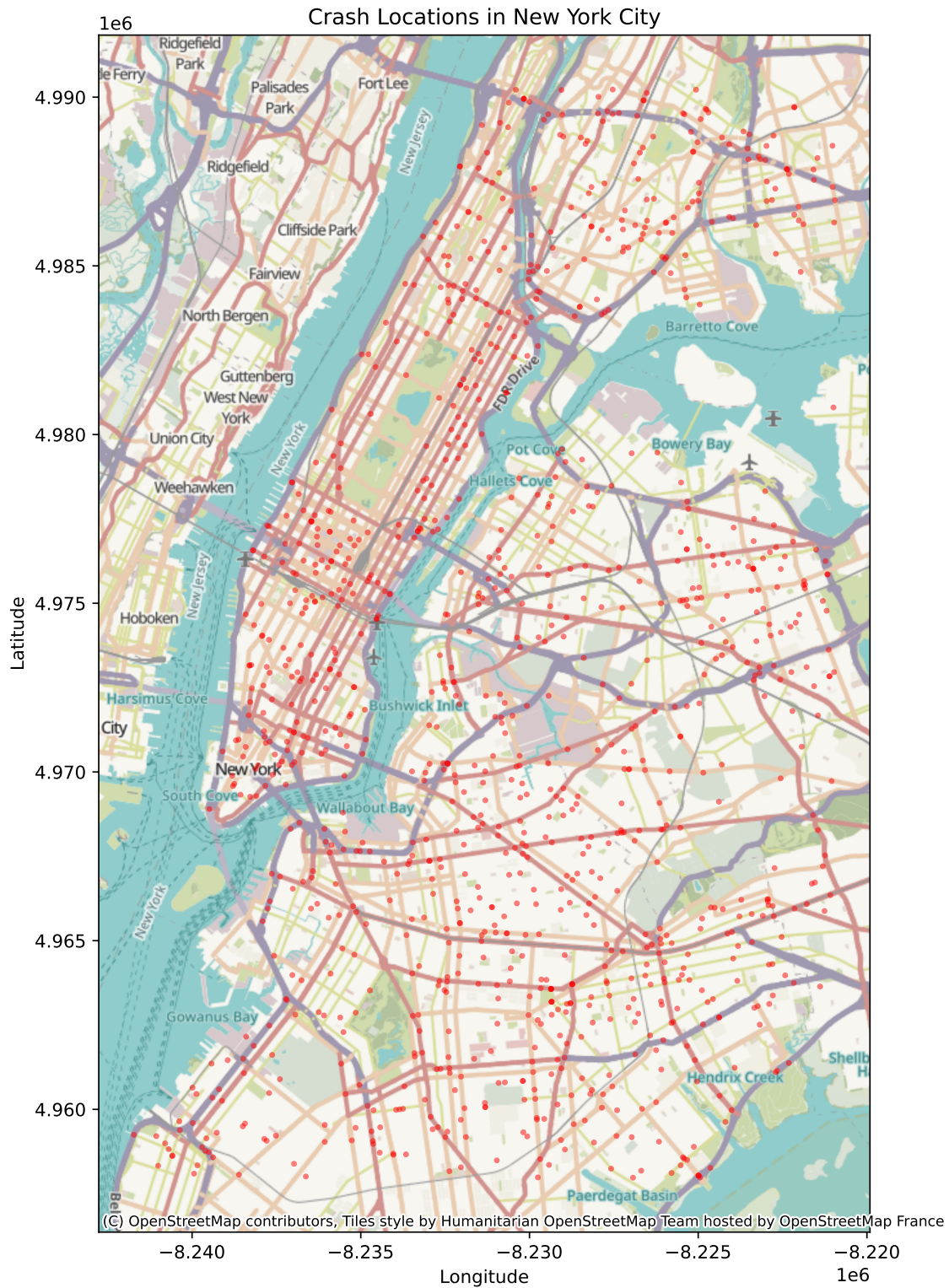
# Create a GeoDataFrame
geometry = [Point(xy) for xy in zip(df['LONGITUDE'], df['LATITUDE'])]
gdf = gpd.GeoDataFrame(df, geometry=geometry)
gdf.crs = "EPSG:4326" # WGS 84

# Filter for NYC bounds (approximate bounds)
nyc_bounds = gdf.cx[-74.05:-73.85, 40.63:40.85]

# Reproject to Web Mercator
gdf_nyc = nyc_bounds.to_crs(epsg=3857)
```

```
# Plotting
fig, ax = plt.subplots(figsize=(12, 12))
gdf_nyc.plot(ax=ax, marker='o', color='red', markersize=5, alpha=0.5)
ctx.add_basemap(ax, crs=gdf_nyc.crs.to_string())

plt.title("Crash Locations in New York City")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.show()
```



4. Create 'Severe' Variable and Construct Cross Table

```
# Create the 'severe' variable
df['severe'] = ((df['NUMBER OF PERSONS INJURED'] > 0) | (df['NUMBER OF PERSONS KILLED'] > 0))

# Crosstab for severity vs borough
severity_borough_table = pd.crosstab(df['BOROUGH'], df['severe'])
print(severity_borough_table)

# Chi-square test
chi2, p, dof, expected = stats.chi2_contingency(severity_borough_table)
print(f"Chi-Square Statistic: {chi2}, P-value: {p}")

if p < 0.05:
    print("Reject the null hypothesis: Severity is associated with borough.")
else:
    print("Fail to reject the null hypothesis: No association between severity and borough.")
```

severe	0	1
BOROUGH		
BRONX	115	98
BROOKLYN	262	200
MANHATTAN	119	109
QUEENS	205	176
STATEN ISLAND	35	15

Chi-Square Statistic: 6.112652162659586, P-value: 0.19089181529657664

Fail to reject the null hypothesis: No association between severity and borough.

5. Merge Crash Data with ZIP Code Database

```
zip_code_data = pd.read_csv('data/nypd_precincts.csv', encoding='latin1') # Wasn't sure what encoding

df['ZIP CODE'] = df['ZIP CODE'].astype(str)
zip_code_data['Precinct'] = zip_code_data['Precinct'].astype(str)

merged_data = pd.merge(df, zip_code_data, left_on='ZIP CODE', right_on='Precinct', how='left')
print(merged_data.head())
```

	CRASH DATE	CRASH TIME	BOROUGH	ZIP CODE	LATITUDE	LONGITUDE	\
0	06/30/2024	1900-01-01 17:30:00	NaN	nan	0.00000	0.00000	
1	06/30/2024	1900-01-01 00:32:00	NaN	nan	NaN	NaN	
2	06/30/2024	1900-01-01 07:05:00	BROOKLYN	11235.0	40.58106	-73.96744	
3	06/30/2024	1900-01-01 20:47:00	NaN	nan	40.76363	-73.95330	
4	06/30/2024	1900-01-01 10:14:00	BROOKLYN	11222.0	40.73046	-73.95149	

	LOCATION	ON STREET NAME	CROSS STREET NAME	\
0	(0.0, 0.0)	NaN	NaN	
1	NaN	BELT PARKWAY RAMP	NaN	
2	(40.58106, -73.96744)	NaN	NaN	
3	(40.76363, -73.9533)	FDR DRIVE	NaN	
4	(40.73046, -73.95149)	GREENPOINT AVENUE	MC GUINNESS BOULEVARD	

	OFF STREET NAME	...	VEHICLE TYPE CODE 2	\
0	GOLD STREET	...	Sedan	
1	NaN	...	Station Wagon/Sport Utility Vehicle	
2	2797 OCEAN PARKWAY	...	NaN	
3	NaN	...	NaN	
4	NaN	...	Box Truck	

	VEHICLE TYPE CODE 3	VEHICLE TYPE CODE 4	VEHICLE TYPE CODE 5	\
0	NaN	NaN	NaN	
1	NaN	NaN	NaN	
2	NaN	NaN	NaN	
3	NaN	NaN	NaN	
4	NaN	NaN	NaN	

	Missing_Geocode	hour	severe	Precinct	Phone	Address
0	False	17	0	NaN	NaN	NaN
1	True	0	0	NaN	NaN	NaN
2	False	7	0	NaN	NaN	NaN
3	False	20	0	NaN	NaN	NaN
4	False	10	0	NaN	NaN	NaN

[5 rows x 35 columns]